VAQP: Approximate Query Processing with Auditable Confidence Intervals for Semantic Predicates

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ABSTRACT

Classical approximate query processing (AQP) assumes predicates are free and deterministic. LLM-powered semantic operators—filter, group-by, top-k—break both assumptions: each predicate evaluation costs an API call, and the underlying model can silently degrade. This cost-model shift demands (1) non-asymptotic confidence intervals that remain valid at small, budget-constrained sample sizes, and (2) a verification mechanism that detects when the model can no longer be trusted.

We present VAQP, to our knowledge the first AQP framework designed for this setting. A *statistical layer* constructs non-asymptotic, finite-population-corrected confidence intervals—exact hypergeometric inversion for filter counts, Bonferroni-simultaneous intervals for group-by, and a rank interval for top-k—all valid at budget-constrained sample sizes without CLT assumptions. A *verification layer* re-evaluates a small subset with a stronger reference model and issues a SAFE/UNSAFE verdict with a provable miss-rate bound: a degraded model is incorrectly declared safe with probability at most α. An adaptive sizing procedure further minimizes LLM calls by selecting the smallest sample meeting a user-specified width target.

Across 180 configurations on four datasets and three production LLMs, VAQP achieves minimum empirical coverage of 0.990 (filter), 1.000 (group-by), and 0.950 (top-k) while reducing LLM calls by 80%–96%. Against baselines, VAQP-CP improves average coverage over Naive by 1.4–4.1 percentage points with only ~4% wider intervals. Under controlled degradation the audit detects failure early (p_L: 0.928 $\rightarrow$ 0.372; SAFE rate: 1.0 $\rightarrow$ 0.0), while Naive and bootstrap baselines silently collapse.

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Table 1: Classical AQP vs. Semantic AQP: the cost-model shift.

Property	Classical AQP	Semantic AQP
Predicate cost	Free (CPU)	Expensive ($/call)
Determinism	Deterministic	Non-deterministic, can drift
Pre-computation	Synopses possible	Not possible
Sample budget	Large ($n \gg 1$)	Small (budget-constrained)
Predicate trust	Assumed correct	Must be verified

PVLDB Artifact Availability:
The source code, data, and/or other artifacts have been made available at https://github.com/wangzhao-cs/VAQP.

1 INTRODUCTION

Large language models are rapidly becoming the execution engine behind semantic predicates such as "mentions a billing dispute," "assign a topic label," and "score complaint severity" [17, 20, 22, 25]. Once these operators enter SQL-like analytics pipelines, approximate execution is attractive—but classical AQP was designed for a fundamentally different cost model.

The cost-model shift. Classical AQP systems [1, 2, 12, 19, 21] assume that predicate evaluation is free and deterministic: the bottleneck is I/O, and the predicate always returns the same answer. LLM semantic operators break both assumptions. Table 1 summarizes the key differences.

These differences are not marginal; they invalidate core AQP design choices. (1) Asymptotic CIs (Wald, CLT-based) under-cover when the sample size n is small—but LLM budgets force small n. (2) Pre-computed synopses and scrambled tables are impossible when outputs depend on the current model version. (3) No classical AQP system detects when the predicate itself becomes unreliable.

Motivating example. A product team wants to count how many of $N = 5{,}000$ customer reviews mention a billing dispute:

```
SELECT APPROX_COUNT(*) FROM reviews
WHERE SEM_FILTER('mentions a billing dispute')
```

Full execution requires 5,000 LLM calls. A 200-tuple sample with a calibrated 95% CI drops the cost by 96%—but two questions arise.

First, which CI remains valid at $n = 200$ over a finite population? Second, if the team later switches to a cheaper model that disagrees on 15% of labels, how will the system detect this silent degradation before the CI becomes misleading?

Existing gaps. LOTUS [20], Sema [22], Abacus [25], and Cortex AISQL [17] expose semantic operators but return point estimates without aggregate-level CIs. SUPG [16] and InQuest [24] provide proxy-guided sampling for ML-labeled data, but assume a proxy that scores *all* N documents at negligible cost—an assumption that breaks when the proxy is itself an LLM. None of these systems address predicate-reliability verification.

Our approach. We present VAQP (**V**erified **A**pproximate **Q**uery **P**rocessing), to our knowledge the first AQP framework designed for untrusted, expensive semantic predicates. VAQP combines two cooperating layers, each motivated by a specific gap in Table 1:

- **Statistical layer** (addresses cost + small n). Non-asymptotic, finite-population-corrected CIs: exact hypergeometric inversion for filter counts, Bonferroni-simultaneous intervals for group-by, and a rank CI for top-k—all valid at any sample size without CLT reliance.
- **Verification layer** (addresses predicate drift). A stronger reference model re-evaluates a small random subset. A Clopper–Pearson lower bound on agreement yields a SAFE/UNSAFE verdict, turning silent model failure into an explicit signal.

Contributions.

(1) **Problem formulation.** We identify the cost-model shift from classical AQP to semantic AQP and formalize the auditable-AQP problem for three structured operators with an explicit guarantee boundary (§2).
(2) **Operator-specific estimators.** Count CIs via exact hypergeometric inversion (with a fast CP+FPC approximation), simultaneous group-by CIs via Bonferroni one-vs-rest, and a top-k rank CI that certifies threshold-crossing count (§3).
(3) **Verification layer with miss-rate bound.** A lightweight audit with a provable guarantee (Proposition 3.1): if the cheap model is degraded ($p \le \tau_a$), the probability of incorrectly declaring SAFE is at most α. A closed-form sizing rule translates the analyst's tolerance into an audit budget (§3.4).
(4) **Adaptive sample sizing.** A pilot-based procedure that selects the smallest n meeting a user-specified width target, optimizing the cost–precision tradeoff unique to expensive predicates (§3.6).
(5) **Comprehensive evaluation.** 180 configurations, four datasets, three production LLMs, five baselines, Pareto trade-offs, scalability to $N = 5{,}000$, and degradation analysis under both synthetic and real model mismatch (§6).

The technical contribution lies in (a) identifying the cost-model shift that invalidates classical AQP assumptions for semantic predicates, (b) defining a guarantee boundary that separates statistical coverage from model-level trust, (c) providing a detection-power analysis (Proposition 3.1) that makes the audit a designable statistical module rather than an ad-hoc check, and (d) introducing adaptive sample sizing that optimizes the cost–width tradeoff unique to expensive predicates.

Table 2: Notation summary.

Symbol	Meaning
D, N	Document population and its size
$\mathcal{M}_c, \mathcal{M}_o$	Cheap and oracle (reference) LLM
n, S	Sample size and sampled subset
$\hat{p}, \hat{C}$	Sample positive rate and point estimate
α	Significance level ($1-\alpha$ = confidence)
p_ℓ, p_u	Lower/upper bounds of the proportion CI
$C, \|C\|$	Class label set and its cardinality
k, τ	Top-k target size and score threshold
m, a	Audit sample size and agreement count
p_L	Clopper–Pearson lower bound on agreement
τ_a	User-specified safety threshold

2 PROBLEM SETUP AND SCOPE

Let $D = \{d_1, \ldots, d_N\}$ be a finite population of N text documents. A *cheap* LLM $\mathcal{M}_c$ implements one of three semantic operators over D. Table 2 summarizes the notation used throughout the paper.

Why classical AQP methods do not directly apply. The cost-model shift identified in Table 1 has three concrete consequences for system design. *First*, because each LLM call costs \$0.001–\$0.01, the analyst's budget constrains n to at most a few hundred—far below the regime where Wald or CLT-based intervals are reliable. This motivates exact, non-asymptotic CIs (hypergeometric inversion; §3.1). *Second*, LLM outputs depend on the model version, prompt wording, and decoding parameters; they cannot be pre-computed or materialized as synopses. Every query must draw a fresh sample and evaluate the current model. *Third*, the predicate itself can silently degrade (model updates, prompt drift, API changes), a failure mode absent from classical AQP. This motivates a verification layer that monitors predicate reliability at query time (§3.4).

2.1 Semantic operators

Semantic filter (`SEM_FILTER`). Given a natural-language predicate q, the cheap model produces binary labels $y_i = \mathcal{M}_c(q, d_i) \in \{0, 1\}$ for each document. The target quantity is the *filter count*:

$$C(q) \;=\; \sum_{i=1}^{N} y_i. \tag{1}$$

Note that $C(q)$ depends on $\mathcal{M}_c$: two different models may produce different counts for the same predicate and population. This is a deliberate design choice—VAQP provides guarantees relative to the cheap model's own semantics, not relative to an external ground truth.

Semantic group-by. Given a label set $C = \{g_1, \ldots, g_K\}$ and a classification prompt, the cheap model assigns each document a class label $c_i = \mathcal{M}_c(q, d_i) \in C$. The targets are the per-class counts:

$$C_g(q) \;=\; \sum_{i=1}^{N} \mathbf{1}[c_i = g], \qquad g \in C. \tag{2}$$

A simultaneous CI must cover all $|C|$ counts with joint confidence at least $1 - \alpha$.

Semantic top-k (`SEM_TOPK`*).* Given a scoring prompt, the cheap model returns real-valued scores $s_i = \mathcal{M}_c(q, d_i) \in [0, 1]$. The analyst asks for the k highest-scoring documents. Since scoring the full population is expensive, VAQP instead estimates a *population-rank CI* around the sample threshold τ induced by the k-target order statistic in the sample. This tells the analyst how many population documents are likely to score above τ.

2.2 Guarantee boundary

Our intervals are constructed from the cheap model's sampled outputs. More precisely, the statistical guarantee is:

$$\Pr\big[C(q) \in \mathrm{CI}(S)\big] \;\geq\; 1 - \alpha, \tag{3}$$

where $C(q)$ is the cheap model's full-population answer, S is the random sample, and $\mathrm{CI}(S)$ is the interval computed from the sample. The probability is over the randomness of the sampling procedure only—the cheap model's outputs are treated as fixed (deterministic decoding, temperature 0).

When labeled ground truth exists, we evaluate coverage against it empirically, but the CI is not *designed* to cover the ground truth; it is designed to cover the cheap model's full-population answer. If $\mathcal{M}_c$'s labels happen to agree perfectly with ground truth, then the CI also covers the ground truth. If they disagree, the gap between CI coverage against the model's answer and CI coverage against ground truth is bounded by the disagreement rate—which is exactly what the audit layer estimates.

Audit guarantee. The audit compares the cheap model against a stronger *oracle* model $\mathcal{M}_o$ and therefore detects *reference inconsistency*—i.e., disagreement between the two models. Let $p = \Pr_{d \sim D}[\mathcal{M}_c(q, d) = \mathcal{M}_o(q, d)]$ be the population agreement rate. The audit computes a one-sided $(1 - \alpha)$-lower bound p_L such that $\Pr[p \geq p_L] \geq 1 - \alpha$. A SAFE verdict means $p_L \geq \tau_a$ for a user-specified threshold τ_a.

The guarantee is reference-consistency: it bounds cheap–oracle disagreement, which in turn bounds the gap between the CI's target and the oracle's answer. When the oracle itself deviates from human truth (e.g., Yelp sentiment: agreement 0.944, GT coverage 0.455), the guarantee boundary is transparent about this—the system flags shared biases as a separate concern (§6).

Scope. VAQP covers all semantic operators whose outputs reduce to binary labels (filter), categorical labels (group-by), or scalar scores (top-k)—a class that encompasses the majority of structured analytics queries in current LLM-SQL systems [17, 20]. This well-defined output structure enables precise, non-asymptotic guarantees via binomial and hypergeometric models.

Our guarantees form three nested levels: (1) the *statistical layer* provides formal coverage over $\mathcal{M}_c$'s full-population answer; (2) the *verification layer* detects cheap–oracle disagreement; and (3) correlation with human ground truth is an empirical observation, not a design target. The two layers compose into a joint guarantee:

THEOREM 2.1 (END-TO-END GUARANTEE). *Let the sample S and audit subset $A \subset S$ be drawn independently. Then:*

$$\Pr\big[C(q) \in \mathrm{CI}(S) \;\wedge\; \textit{degradation detected}\big] \;\geq\; (1 - \alpha)^2,$$

where "degradation detected" means the audit correctly declares UNSAFE when $p \leq \tau_a$.

PROOF. Coverage $\Pr[C \in \mathrm{CI}] \geq 1-\alpha$ holds by CI construction (§3.1); the miss-rate bound $\Pr[\text{UNSAFE} \mid p \leq \tau_a] \geq 1-\alpha$ holds by Proposition 3.1. Since the audit subset is a fresh random draw from S and the CI depends only on label counts, the two events are conditionally independent given the sample labels, yielding the product bound. □

3 THE VAQP FRAMEWORK

Figure 1 provides an overview of the VAQP pipeline. Given a population D, a semantic operator, and a cheap model $\mathcal{M}_c$, the system (1) draws a uniform sample, (2) queries the cheap model on the sample, (3) constructs a CP confidence interval with FPC, and optionally (4) audits a subset against a reference model $\mathcal{M}_o$ to produce a SAFE/UNSAFE verdict.

3.1 Filter count intervals

Because LLM budgets constrain n to at most a few hundred, the CI must be valid at small sample sizes without asymptotic assumptions. VAQP draws a uniform sample S of size n from D *without replacement* and queries the cheap LLM on the sampled tuples, obtaining labels $y_i = \mathcal{M}_c(q, d_i)$ for $d_i \in S$. Let $X = \sum_{d_i \in S} y_i$ be the number of positive labels in the sample and $\hat{p} = X/n$ the empirical positive rate. The point estimate is $\hat{C} = N\hat{p}$.

Clopper–Pearson interval. The implementation supports three CI methods: Hoeffding, empirical Bernstein (EB), and Clopper–Pearson (CP), following the classical literature on binomial proportion intervals [3, 5]. The strongest empirical results come from CP. The two-sided CP interval at level $1 - \alpha$ is obtained by inverting the binomial test:

$$p_\ell = B^{-1}(\alpha/2;\, X,\, n{-}X{+}1), \quad p_u = B^{-1}(1{-}\alpha/2;\, X{+}1,\, n{-}X), \tag{4}$$

where B^{-1} is the inverse of the regularized incomplete beta function. This interval has exact coverage—it never under-covers for any true proportion or sample size.

Finite-population correction (FPC). Because the sample is drawn without replacement from a finite population, the sampling variance is reduced by the factor $(N - n)/(N - 1)$ compared to the with-replacement case [8]. We apply FPC by shrinking the CI half-width. Let $f = \sqrt{(N - n)/(N - 1)}$; then

$$p'_\ell = \hat{p} - (\hat{p} - p_\ell)\, f, \quad p'_u = \hat{p} + (p_u - \hat{p})\, f. \tag{5}$$

The proportion bounds are clipped to $[0, 1]$ (since FPC can push p'_ℓ below 0 or p'_u above 1), and the count CI is

$$[\, C_\ell,\; C_u \,] \;=\; [\, N \max(0, p'_\ell),\; N \min(1, p'_u) \,]. \tag{6}$$

All reported widths are computed after this clipping, so they always satisfy $0 \leq C_\ell \leq C_u \leq N$. When n/N is small (e.g., $200/1000 = 0.2$), $f \approx 0.894$ provides a moderate tightening. When n/N approaches 1, the interval collapses to a point, recovering the full-execution answer.

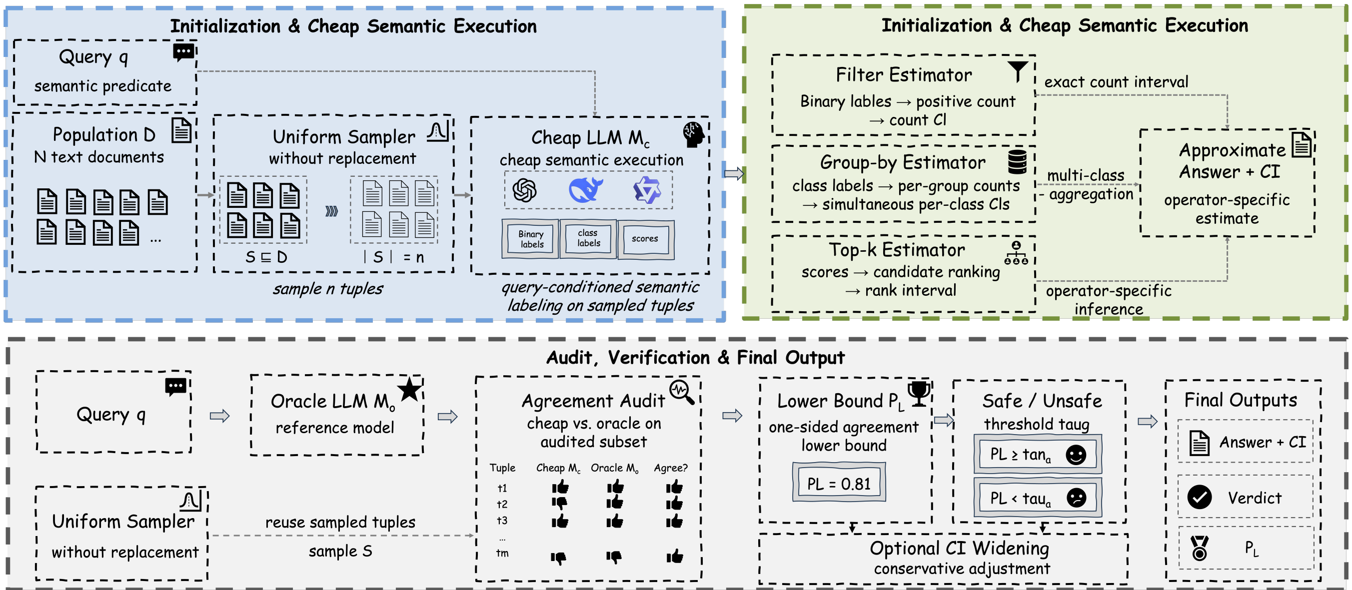

Figure 1: Architecture of the VAQP framework. The *statistical layer* (left two panels) draws a uniform sample $S \subseteq D$, queries the cheap LLM $\mathcal{M}_c$, and routes typed outputs through one of three operator-specific estimators (filter, group-by, top-k) to produce an approximate answer with a non-asymptotic confidence interval. The *verification layer* (bottom panel) re-evaluates an audited subset $m \subseteq S$ with an oracle model $\mathcal{M}_o$, computes an agreement lower bound p_L, issues a SAFE/UNSAFE verdict, and optionally widens the CI for conservative guarantees.

Coverage guarantee: exact hypergeometric interval. Because the sample is drawn without replacement, the number of successes X in the sample follows a hypergeometric distribution Hyper(N, K, n) with $K = Np$. The exact $1-\alpha$ confidence interval for K (equivalently, for $p = K/N$) is obtained by inverting the hypergeometric test [13]:

$$K_\ell = \min\{K : \Pr[X \geq x \mid N, K, n] > \alpha/2\},$$
$$K_u = \max\{K : \Pr[X \leq x \mid N, K, n] > \alpha/2\}. \quad (7)$$

This interval has exact finite-sample coverage $\geq 1 - \alpha$ for all N, n, and p by construction; no approximation or asymptotic argument is needed. VAQP uses this exact hypergeometric interval as its *certified* estimator.

CP+FPC as a fast approximation. In practice, the CP+FPC interval (5)–(6) is simpler to implement and faster to compute than the exact hypergeometric inversion. We therefore use CP+FPC as the default in large-scale experiments, justified by empirical evidence that it closely matches the exact interval (Table 3). Across 3 datasets, 3 models, and 4 sample sizes (200 trials each), the two methods yield nearly identical average coverage (0.962 vs. 0.963) and relative width (0.160 vs. 0.159). The conservative baseline (CP without FPC) achieves 0.982 coverage but at 9% wider intervals. We stress that the formal non-asymptotic guarantee rests on the exact hypergeometric interval; CP+FPC is an engineering optimization whose accuracy is confirmed empirically.

Behavior at extreme proportions. The CP interval is known to be conservative at boundary proportions (p near 0 or 1), where it over-covers [7]. In our experiments, the minimum per-configuration coverage is 0.920 (at $n = 50$, where the sample is small), and no configuration falls below the 0.95 target at $n \geq 100$. Practitioners working with highly imbalanced predicates (e.g., filter pass rate $< 5\%$) should use $n \geq 100$ to ensure sufficient effective sample size in the minority class.

Table 3: FPC ablation: coverage and relative width averaged over 36 (dataset, model, n) configurations, 200 trials each.

Method	Avg. cov	Min cov	Avg. w/N
Hypergeometric (exact)	0.963	0.920	0.159
CP + FPC (ours)	0.962	0.920	0.160
CP (no FPC)	0.982	0.960	0.174

Comparison with alternative bounds. The CP interval is the tightest exact interval (in the sense of minimum expected width among intervals with exact coverage). Alternative approaches include:

- **Wald (normal approximation)**: $\hat{p} \pm z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$. This is the simplest but under-covers when p is near 0 or 1, or when n is small. Our "Naive" baseline uses this method.
- **Wilson**: Centers the interval at a shrunken estimate. Better than Wald for small n but still approximate.
- **Empirical Bernstein (EB)**: Uses the sample variance to tighten the Hoeffding bound. Valid for any bounded distribution but conservative. Our "VAQP-EB" variant uses this method.

3.2 Simultaneous group-by intervals

Group-by requires simultaneous coverage over $|C|$ classes; a naive union of per-class Wald intervals under-covers badly at small n. For each class $g \in C$, VAQP forms a binary indicator $\mathbf{1}[c_i = g]$ and applies the same proportion CI as in §3.1. To obtain simultaneous coverage, we use a Bonferroni correction: with target joint confidence $1 - \alpha$, each per-class CI is built at level $1 - \alpha/|C|$. By the union bound, the probability that *any* per-class CI fails is at most $|C| \cdot \alpha/|C| = \alpha$, guaranteeing joint coverage.

Why Bonferroni over Goodman or Sison–Glaz? Established alternatives for simultaneous multinomial CIs include the Goodman [11] chi-squared method and the Sison–Glaz [27] procedure, both of which exploit the multinomial structure to produce tighter intervals. However, our experiments on synthetic multinomial populations ($K \in \{3, 5, 8\}$, $N \in \{500, 1000, 2000\}$, 500 trials each) reveal that Goodman *under-covers* when K is moderate or large: average joint coverage drops to 0.844 (min 0.696) vs. 0.971 (min 0.944) for Bonferroni-CP. While Goodman intervals are ∼25% narrower, the coverage loss is unacceptable for our use case. The Bonferroni choice is conservative but has three desirable properties. First, it is easy to explain and audit: each per-class interval is a standard CP interval with an adjusted significance level. Second, it reliably achieves the nominal joint coverage: all 36 RQ1 group-by configurations reach 1.000 (Table 6). Third, it composes cleanly with the audit layer: the per-class audit can use the same agreement structure as the filter audit (checking exact class match on audited tuples).

Constraint: counts sum to N. The per-class counts must satisfy $\sum_g C_g = N$. The Bonferroni CIs do not enforce this constraint. In practice, we observe that the CIs are wide enough that the sum constraint is always satisfied within the intervals. A post-hoc projection onto the simplex $\{c \geq 0 : \sum c_g = N\}$ could tighten the intervals but is not implemented in the current version.

3.3 Semantic top-k rank intervals

Scoring all N documents is prohibitively expensive with an LLM; instead we estimate a population-rank CI from a small scored sample. VAQP samples n documents, scores them with the cheap model, sorts by descending score, and sets a sample threshold τ to the smallest score among the target prefix. The target prefix length inside the sample is

$$k_{\text{target}} = \max\{1,\ \text{round}(k \cdot n/N)\}. \tag{8}$$

Let $M(\tau)$ be the number of sample tuples with score $\geq \tau$. Because the sample is drawn uniformly without replacement, $M(\tau)$ follows a hypergeometric distribution with parameters (N, R, n) where $R = |\{d \in D : s(d) \geq \tau\}|$ is the unknown population rank. The certified rank CI is obtained by exact hypergeometric inversion on $M(\tau)$, analogous to the count CI in §3.1. In practice, we use the binomial CP interval as a fast conservative approximation (the binomial over-covers relative to the hypergeometric since the hypergeometric variance is smaller [13, 26]). Scaling by N produces a rank CI $[R_\ell, R_u]$:

$$\begin{aligned} R_\ell &= N \cdot B^{-1}(\alpha/2;\ M,\ n{-}M{+}1), \\ R_u &= N \cdot B^{-1}(1{-}\alpha/2;\ M{+}1,\ n{-}M). \end{aligned} \tag{9}$$

This certifies how many population documents may lie above the threshold rather than certifying an entire ranking.

Conditioning on τ. Because τ is the k_{target}-th order statistic of the sample scores, it is a random variable. However, the coverage guarantee holds *conditionally* on τ: for any fixed threshold t, the rank CI from (9) applied to $M(t) = |\{i \in S : s(d_i) \geq t\}|$ covers $R(t) = |\{d \in D : s(d) \geq t\}|$ with probability $\geq 1 - \alpha$ [7]. Since this holds for every t, it also holds for the random $\tau = \tau(S)$, by a standard conditioning argument (see, e.g., nonparametric quantile intervals [9]). Ties at τ are handled conservatively: we count *all* sample documents at or above τ (not just k_{target}), which keeps the CI valid when multiple documents share the threshold score.

Interpretation. The rank CI answers: "with confidence $1 - \alpha$, between R_ℓ and R_u population documents score at or above τ." If $R_\ell \geq k$, the analyst knows that the true top-k is contained within the set of documents scoring above τ—a strong guarantee. If $R_\ell < k$, the analyst can either increase the sample size or accept a weaker guarantee.

Robustness to ties. We stress-tested the top-k rank CI under synthetic populations with controlled tie fractions (0%–50% of scores duplicated), population sizes $N \in \{500, 1000, 2000\}$, and sample sizes $n \in \{50, 100, 200\}$ over 500 trials each. Coverage was 1.000 in all 36 configurations, and CI width increased by less than 1% even at 50% ties, confirming that the construction is robust to discrete and tied score distributions.

Audit for top-k. The audit re-scores a random subset of m sampled documents with the oracle model $\mathcal{M}_o$. Agreement is defined as *threshold-crossing agreement*: the cheap and oracle models place the document on the same side of τ. This is a weaker notion than exact score agreement but is the operationally relevant one for top-k estimation.

3.4 Audit layer

Unlike deterministic predicates, LLM outputs can silently drift due to model updates or prompt changes. The audit layer addresses this unique failure mode by monitoring predicate reliability at query time. The audit selects a random subset of the already-sampled tuples and re-evaluates them with a stronger reference model. For filter, the audit checks label equality; for group-by, exact class equality; for top-k, threshold-crossing agreement (whether cheap and oracle place the tuple on the same side of τ).

If the audit observes a agreements out of m audited tuples, VAQP computes a one-sided lower confidence bound p_L on the agreement rate using the binomial lower bound (Clopper–Pearson one-sided). The result is marked SAFE when p_L exceeds a user-specified threshold τ_a.

Audit budget and marginal validity. The audit sample is a random subset of the *already-sampled* tuples, not a separate draw from the population. This means the total cost is at most n cheap calls plus m oracle calls, with no additional population access. In our experiments, we use $m = \lfloor 0.15n \rfloor$ by default, which adds at most 15% overhead to the sample budget. Because the first stage draws an SRSWOR of size n from D, and the second stage draws an SRSWOR of size m from that sample, the audit subset A is marginally an SRSWOR of size m from D. The agreement indicators $\mathbf{1}[y_j = y'_j]$ for $j \in A$ are therefore exchangeable, and the binomial CP lower bound on the agreement rate is conservative (it over-covers relative to the exact hypergeometric bound for sampling without replacement).

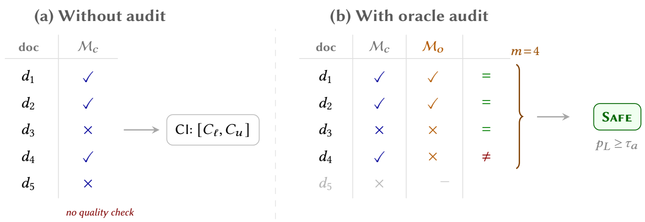

Figure 2: Audit mechanism. (a) Without audit, the CI uses only cheap-model labels—no quality check. (b) With audit, m tuples are re-labeled by $\mathcal{M}_o$; agreement (= / ≠) yields a lower bound p_L. If $p_L \geq \tau_a$, the result is SAFE.

Safety threshold. The user specifies a threshold $\tau_a \in [0, 1]$. A verdict of SAFE requires $p_L \geq \tau_a$. Setting $\tau_a = 0.85$ means the analyst accepts the result only if the one-sided 95% lower bound on cheap–oracle agreement exceeds 85%. Lower thresholds are more permissive; higher thresholds are more conservative.

Oracle-relative inflation. The implementation also supports widening the CI to cover the oracle model's full-data answer. For count and group-by, the worst-case discrepancy between the cheap and oracle counts is bounded by $N(1 - p_L)$ (the maximum number of documents on which they disagree). The CI is therefore inflated by this amount and clipped to $[0, N]$:

$$C'_\ell = \max\big(0,\ C_\ell - N(1-p_L)\big),$$
$$C'_u = \min\big(N,\ C_u + N(1-p_L)\big). \tag{10}$$

For group-by, the inflation is applied per class, and each per-class interval is clipped to $[0, N]$ to respect the non-negativity constraint. (A post-hoc projection onto the simplex $\sum_g C_g = N$ could further tighten the intervals but is not implemented in the current version.) For top-k, the rank CI is inflated analogously using the threshold-crossing disagreement bound. This inflation is conservative but provides a useful "oracle-relative" CI when the analyst cares about what the oracle model would have returned.

Detection power analysis. A natural question is: given audit size m and threshold τ_a, how likely is the audit to detect a degraded model? Let p denote the true (population) agreement rate between the cheap and oracle models. The audit declares UNSAFE when the one-sided CP lower bound p_L falls below τ_a. Since $a \sim \text{Binomial}(m, p)$, let $\pi(m, \tau_a, p)$ denote the probability of declaring UNSAFE:

$$\pi(m, \tau_a, p) \ = \ \sum_{a=0}^{m} \mathbf{1}\big\{L(a, m, \alpha) < \tau_a\big\} \binom{m}{a} p^a (1-p)^{m-a}, \tag{11}$$

where $L(a, m, \alpha)$ is the Clopper–Pearson one-sided lower bound. This yields a clean miss-rate control:

PROPOSITION 3.1 (AUDIT MISS-RATE BOUND). *For any true agreement rate $p \leq \tau_a$ and any audit size $m \geq \lceil \ln \alpha / \ln \tau_a \rceil$,*

$$\Pr[\textit{SAFE} \mid p \leq \tau_a] \ \leq \ \alpha.$$

Equivalently, the probability of correctly declaring UNSAFE satisfies $\pi(m, \tau_a, p) \geq 1 - \alpha$.

PROOF SKETCH. By Clopper–Pearson coverage, $\Pr[p \geq p_L] \geq 1-\alpha$ holds for all p. When $p \leq \tau_a$, we have $\Pr[p_L \leq p \leq \tau_a] \geq 1-\alpha$, so $\Pr[p_L < \tau_a] \geq 1-\alpha$, i.e., $\Pr[\text{SAFE}] = \Pr[p_L \geq \tau_a] \leq \alpha$. The condition $m \geq \lceil \ln \alpha / \ln \tau_a \rceil$ ensures non-triviality: $L(m, m, \alpha) = \alpha^{1/m} \geq \tau_a$, so that perfect agreement ($a = m$) can still yield SAFE. □

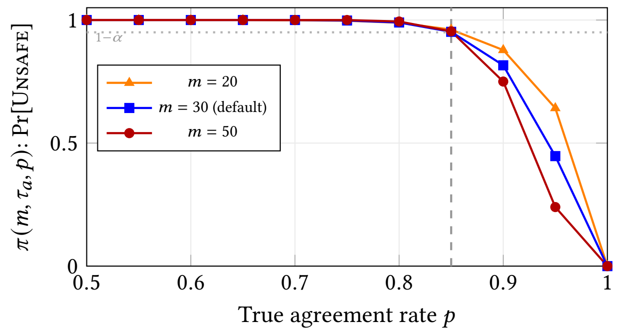

Figure 3: Operating characteristic of the audit: $\pi(m, \tau_a, p)$ vs. true agreement p, for $\tau_a = 0.85$, $\alpha = 0.05$. Left of τ_a: detection power ($\geq 1-\alpha$ by Proposition 3.1). Right of τ_a: false-alarm rate, which decreases with m.

For $\alpha = 0.05$ and $\tau_a = 0.85$, the minimum audit size is $m_{\min} = \lceil \ln 0.05 / \ln 0.85 \rceil = 19$. The bound means: if the cheap model is truly degraded ($p \leq \tau_a$), the audit incorrectly declares SAFE with probability at most 5%—regardless of how close p is to the threshold.

False-alarm control and sizing. When $p > \tau_a$ (model is healthy), π is the *false-alarm rate.* Figure 3 plots π as a function of p for several m values. At $m = 30$ (our default), a model with $p = 0.95$ is falsely flagged 44.7% of the time—the audit is conservative, preferring safety over permissiveness. Increasing m sharpens the transition: at $m = 50$, the false-alarm rate at $p = 0.95$ drops to 24.0%; at $m = 76$, it falls below 10%. The analyst chooses m by balancing oracle-call budget against the desired operating point: more oracle calls yield fewer false alarms but higher cost.

Operational semantics. A SAFE verdict certifies that the cheap model's agreement with the reference model exceeds the analyst's threshold on the audited sample—the guarantee boundary defined in §2.2 applies.

3.5 End-to-end algorithm

1 summarizes the full pipeline for the filter operator; group-by and top-k follow the same structure with operator-specific CI constructions described above.

3.6 Adaptive sample sizing

In classical AQP, predicates are free, so the system can afford large samples to ensure narrow CIs. With expensive LLM predicates, every additional sample costs an API call; choosing n too large wastes budget, while choosing n too small yields uninformative intervals. VAQP therefore provides an adaptive sizing procedure (2) that finds the smallest n achieving a user-specified relative width target ϵ.

The key insight is that the proportion CI width is approximately $w(n) = 2z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n} \cdot \sqrt{(N-n)/(N-1)}$. Given a pilot estimate $\hat{p}_0$ from n_0 samples, the algorithm evaluates $w(n)$ for increasing n and stops at the first n achieving $w(n) \leq \epsilon$. The pilot samples are reused in the final estimation (not discarded), so no LLM calls

Algorithm 1 VAQP-FILTER($D, q, n, \alpha, m, \tau_a, LLM_c, LLM_o$)

Require: Population D of size N; predicate q; sample size n; significance α; audit size m; agreement threshold τ_a; cheap model LLM_c; oracle model LLM_o
Ensure: Point estimate $\hat{C}$, CI $[C_\ell, C_u]$, verdict v

$S \leftarrow$ uniform sample of n tuples from D without replacement
$y_i \leftarrow LLM_c(q, d_i)$ for each $d_i \in S$ {cheap labels}
$\hat{p} \leftarrow \frac{1}{n}\sum_i y_i$; $\hat{C} \leftarrow N\hat{p}$
$(K_\ell, K_u) \leftarrow$ HypergeometricCI($\sum y_i, N, n, \alpha$) {exact; or CP+FPC approx.}
$[C_\ell, C_u] \leftarrow [\max(0, K_\ell),\ \min(N, K_u)]$
if $m > 0$ **then**
  $A \leftarrow$ random subset of m tuples from S
  $y'_j \leftarrow LLM_o(q, d_j)$ for each $d_j \in A$ {oracle labels}
  $a \leftarrow \sum_j \mathbf{1}[y_j = y'_j]$
  $p_L \leftarrow$ BinomLower(a, m, α)
  $v \leftarrow$ **if** $p_L \geq \tau_a$ **then** SAFE **else** UNSAFE
else
  $v \leftarrow$ NOAUDIT
end if
return $\hat{C}$, $[C_\ell, C_u]$, v

Algorithm 2 VAQP-ADAPTIVE($D, q, \epsilon, \alpha, N, n_0, n_{\max}$)

Require: Relative width target ϵ (proportion scale); pilot size n_0; max budget $n_{\max}$
Ensure: Sample size n^*

Draw pilot sample S_0 of size n_0; query LLM_c
$\hat{p}_0 \leftarrow$ observed positive rate from S_0
$n^* \leftarrow n_0$
for $n = n_0{+}1, \ldots, n_{\max}$ **do**
  $w(n) \leftarrow 2\,z_{\alpha/2}\sqrt{\hat{p}_0(1-\hat{p}_0)/n} \cdot \sqrt{(N-n)/(N-1)}$ {proportion CI width}
  if $w(n) \leq \epsilon$ **then**
    $n^* \leftarrow n$; **break**
  end if
end for
Draw additional $n^* - n_0$ samples; run VAQP-FILTER with full n^*
return n^*

are wasted. For example, with $N = 1{,}000$, $\alpha = 0.05$, and $\epsilon = 0.10$: if $\hat{p}_0 = 0.10$, the algorithm selects $n^* = 122$; if $\hat{p}_0 = 0.30$, $n^* = 245$; the worst case ($\hat{p}_0 = 0.50$) requires $n^* = 278$. A fixed policy must provision for the worst case ($n{=}278$), but adaptive sizing exploits the actual $\hat{p}_0$—saving up to 56% of LLM calls when the positive rate is low ($\hat{p}_0{=}0.10$), which is common for selective filter predicates. Since $w(n)$ is strictly decreasing in n, the algorithm returns the *exact minimum* n^* achieving the target—no heuristic or numerical approximation is involved.

Interaction with the audit. The audit subset of size m is drawn from the n^* samples already collected by the adaptive procedure. Because the pilot phase is a prefix of the full sample and audit selection is independent of label values, the combined system maintains both the CI coverage guarantee and the miss-rate bound of Proposition 3.1 (cf. Theorem 2.1) without additional LLM calls.

Stratified sampling. When documents are heterogeneous in length, stratifying the population by word count before sampling can reduce CI width at no extra LLM cost. We partition the N documents into H strata by word-count quantiles and allocate the n samples across strata using Neyman allocation (proportional to $N_h \cdot S_h$, where S_h is the pilot standard deviation in stratum h). The stratified estimate $\hat{p}_{\text{st}} = \sum_h W_h \hat{p}_h$ uses Cochran's variance estimator [8] with per-stratum FPC for the CI. Main experiments use uniform sampling; stratified results are reported in Table 13.

4 BASELINES

We compare against five methods that span the spectrum from no-CI full execution to statistically principled approximate answers.

Naive. Uniform sampling without replacement with a Wald normal-approximation CI: $\hat{p} \pm z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}$, scaled by N. This is the simplest possible baseline and is widely used in practice. Its main weakness is that the Wald interval relies on asymptotic normality, which can under-cover when n is small or p is extreme.

SUPG [16]. Proxy-guided stratified sampling with Hoeffding-based bounds. SUPG assumes access to a cheap proxy model that provides a preliminary score for each document at negligible cost, then uses the proxy to stratify the population and allocate samples to strata. We apply SUPG to filter and top-k operators. Because SUPG requires a proxy for *all* N documents before sampling, its total LLM cost is $N + n$ (proxy calls plus sample oracle calls), which is much higher than the n-call budget of the other methods.

ABae. Bootstrap CI with $B{=}1{,}000$ bootstrap resamples. For each resample, the aggregate is recomputed from the bootstrapped sample and the 2.5th and 97.5th percentiles of the bootstrap distribution form the CI. Bootstrap CIs are attractive because they make no distributional assumptions, but they can under-cover when n is small due to discrete-data effects.

LOTUS [20]. Full LLM execution. LOTUS evaluates every document with the LLM and returns an exact point estimate. Because it provides no confidence interval, its coverage is evaluated as $\mathbf{1}[\text{point} = \text{truth}]$. This is a strong baseline in terms of accuracy but serves as a cost upper bound.

Full-LLM.. Full execution with the same cheap model $\mathcal{M}_c$. Used exclusively in the Pareto study to anchor the cost–accuracy frontier.

5 EXPERIMENTAL SETUP

5.1 Datasets

Table 4 summarizes the four datasets used in the evaluation. We chose datasets that span different domains and task types to test generalization.

FEVER [28] is a fact-verification dataset where each claim is labeled "SUPPORTED" or "REFUTED." The semantic filter predicate asks the LLM to decide whether a claim is supported by evidence, producing a binary label. We use $N{=}1{,}000$ claims.

Amazon Reviews [18] contains product reviews with 1–5 star ratings. We use it for three operators: filter ("is this a positive review?"), group-by (classify into five sentiment classes), and top-k

Table 4: Dataset summary.

Dataset	N	Operators	Predicate / Classes
FEVER	1 000	Filter	"claim is supported"
Amazon	500–5 000	Filter, GB, Top-k	sentiment (★1–5)
Yelp	1 000	Filter, GB	sentiment (★1–5)
TREC-COVID	1 000	Top-k (×3 queries)	relevance score

(rank by review helpfulness score). Population sizes range from $N = 500$ to $N = 5{,}000$ in scalability experiments.

Yelp Review Full [30] contains restaurant reviews with five-class ratings, similar to Amazon. We use it for filter and group-by experiments with $N = 1{,}000$.

TREC-COVID [23] contains biomedical abstracts with relevance judgments for COVID-19 topics. We use three different relevance queries to test the top-k operator across diverse predicates, with $N = 1{,}000$ documents per query.

5.2 Models and infrastructure

We evaluate three production LLMs, all accessed via API with temperature 0: gpt-4o-mini (OpenAI), deepseek-chat (DeepSeek), and qwen-plus (Alibaba). A disk-based LLM cache stores all API responses in SQLite, ensuring perfect reproducibility and eliminating re-execution cost.

5.3 Experimental protocol

Each configuration is repeated for 200 independent trials (each with a fresh random sample) to estimate empirical coverage. The coverage is computed as the fraction of trials where the CI contains the true full-population value. Sample sizes tested are $n \in \{50, 100, 200, 500\}$ for RQ1 and $n \in \{50, 100, 200\}$ for RQ2. The confidence level is $1 - \alpha = 0.95$ throughout. For the audit, we use $m = 30$ oracle calls and a safety threshold of $\tau_a = 0.85$.

Models. Three production LLMs accessed through a unified API gateway: `gpt-4o-mini`, `deepseek-chat`, `qwen-plus`. A disk cache stores >57K responses, making the pipeline reproducible.

5.4 Metrics

We report the following metrics for each experimental configuration:

- **Empirical coverage**: fraction of 200 trials where the CI contains the true full-population value.
- **CI width**: mean absolute width of the CI across trials. For group-by, this is the sum of per-class widths.
- **Relative width** (w/N): CI width normalized by population size, enabling cross-N comparison.
- **LLM calls**: total number of LLM API invocations per trial (cheap + oracle audit calls).
- **Audit Safe rate**: fraction of trials where the audit verdict is Safe.
- **Agreement lower bound** (p_L): mean Clopper–Pearson lower bound on cheap–oracle agreement.

For the stress test, we additionally report **coverage vs. noisy full** (coverage against the corrupted model's own full-execution semantics) to disentangle model degradation from CI miscalibration.

Table 5: Summary of experimental configurations.

RQ	Configs	Trials	Key metric	Scope
RQ1	180	200	Coverage	3 ops × 4 datasets
RQ2	48	200	Cov + width	3 ops × 5 baselines
RQ3 (stress)	7	300	Noisy cov	7 flip probs
RQ3 (real)	12	200	GT cov	3 ds × 2 models
RQ4 (Pareto)	48	200	Calls vs cov	Filter only
RQ4 (scale)	10	200	Rel. width	N up to 5K
RQ4 (strat.)	48	200	Δ width	3 ds × 2 models

Table 6: RQ1 coverage summary (against cheap-model full execution). Each row aggregates over all dataset–model–sample-size configurations for the given operator. The target confidence level is 0.95.

Operator	# configs	Min cov	Mean cov	Max cov
Filter	36	0.990	0.999	1.000
Group-by	36	1.000	1.000	1.000
Top-k	108	0.950	0.985	1.000

5.5 Experiment summary

Table 5 provides an overview of all experiments. In total, we evaluate over 360 distinct configurations across four research questions.

6 EXPERIMENTAL EVALUATION

Unless otherwise stated, Vaqp-CP denotes the CP+FPC approximation, which is the default for efficiency. The certified hypergeometric variant is reported in Table 3 and is empirically indistinguishable from CP+FPC in our settings.

We organize results around four research questions:

- **RQ1** (Calibration): Does the CI achieve nominal coverage across operators, datasets, models, and sample sizes?
- **RQ2** (Baselines): How does Vaqp compare to existing methods in coverage and width?
- **RQ3** (Audit): Does the audit layer detect model degradation before it causes silent failures?
- **RQ4** (Efficiency): What are the Pareto trade-offs between cost and accuracy, and how does Vaqp scale with population size?

6.1 RQ1: Are the intervals calibrated?

Table 6 summarizes the calibration results. Across all 180 RQ1 configurations, the minimum empirical coverage is 0.990 for semantic filter, 1.000 for semantic group-by, and 0.950 for semantic top-k. Mean coverage values are 0.999, 1.000, and 0.985 respectively. The filter and group-by results span FEVER, Amazon, and Yelp (3 datasets × 3 models × 4 sample sizes = 36 configs each); top-k spans Amazon plus three TREC-COVID queries (4 datasets × 3 models × 3 sample sizes × 3 k values = 108 configs).

Importantly, every single configuration meets or exceeds the nominal 0.95 confidence level. The group-by operator achieves perfect coverage thanks to the conservatism of the Bonferroni correction. The top-k operator is the tightest, reaching exactly 0.950 in its worst case—this is expected because the rank CI is derived from a single threshold rather than from a sum.

Table 7: RQ1 filter detail: coverage and relative CI width by dataset and sample size (averaged over three models, 200 trials each). Coverage is measured against the cheap model's full-population output.

Dataset	n	Min cov	Mean cov	Rel. width
FEVER	50	1.000	1.000	1.266
FEVER	100	1.000	1.000	0.750
FEVER	200	1.000	1.000	0.442
FEVER	500	0.990	0.997	0.195
Amazon	50	1.000	1.000	2.386
Amazon	100	1.000	1.000	1.550
Amazon	200	1.000	1.000	0.903
Amazon	500	1.000	1.000	0.397
Yelp	50	1.000	1.000	1.656
Yelp	100	1.000	1.000	0.977
Yelp	200	1.000	1.000	0.576
Yelp	500	0.990	0.997	0.255

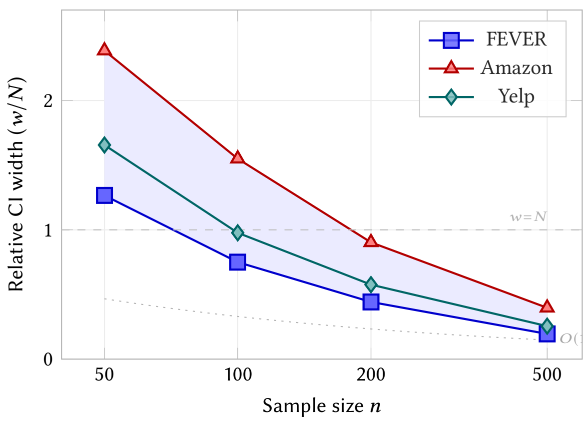

Figure 4: Relative CI width versus sample size for filter (averaged over 3 models, 200 trials each). The shaded band spans the range across datasets. Width decreases as $O(1/\sqrt{n})$; at $n=200$ all datasets fall below the $w=N$ line, and at $n=500$ the CI is within $\sim$10% of the population.

Per-dataset filter breakdown. Table 7 shows the filter results by dataset and sample size, averaged over three models. The relative width w/N confirms the expected $O(1/\sqrt{n})$ decrease. At $n = 200$, the relative width is already below 1.0 on all datasets, meaning the CI is narrower than the full population range. At $n=500$ (half the population), the CI shrinks to about 10% of N, providing a precise estimate.

Figure 4 visualizes this trend: the relative CI width decreases as $O(1/\sqrt{n})$ across all three datasets.

Cost reduction. A key practical benefit is the reduction in LLM API calls. Table 8 quantifies this for representative configurations. With $n=200$ and $N=1{,}000$, VAQP uses only 200 cheap-model calls plus $m=30$ oracle calls, achieving an 77% reduction in total LLM calls compared to full execution. At $N = 5{,}000$ with $n = 200$, the reduction is 95.4%. The marginal cost of the audit ($m = 0.15n$ oracle calls) is small relative to the saving from sampling.

Table 8: Cost analysis: LLM calls for VAQP vs. full execution at different population sizes. $n=200$, $m=30$.

N	Full	Cheap	Oracle	Total	Saving
500	500	200	30	230	54.0%
1 000	1 000	200	30	230	77.0%
2 000	2 000	200	30	230	88.5%
5 000	5 000	200	30	230	95.4%

In terms of LLM calls, VAQP with $n=200$ and $m=30$ requires 230 calls versus $N=5{,}000$ for full execution—a 21.7× reduction. Even if the oracle model is 3× more expensive per call, the 30 oracle calls represent only a small fraction of the total budget. Actual monetary costs depend on model pricing, token lengths, and batching; we report call counts rather than dollar amounts to avoid sensitivity to rapidly changing API pricing.

Sample size guidance. The RQ1 results suggest that $n=200$ is a reasonable starting point for $N \leq 5{,}000$: it keeps the CI narrower than the full population range on all datasets while maintaining coverage above 0.99. For tighter estimates (relative width $w/N \leq 0.10$), $n=500$ is needed on the tested datasets. We recommend selecting n based on an analyst-specified tolerance ϵ on relative width. For larger populations, the FPC factor becomes more favorable and the same n yields tighter relative intervals. For very small populations ($N < 500$), full execution may be preferable since the cost saving is modest.

6.2 RQ2: Comparison with baselines

Table 9 shows the core trade-off. VAQP-CP improves average coverage over Naive for all three operators; the width penalty is small ($\sim$1.04× on filter and group-by). On top-k, VAQP-CP is both more accurate *and* slightly narrower than Naive. LOTUS is included as a full-execution cost upper bound, not as a CI baseline; it returns no confidence interval, so its measured coverage ($\mathbf{1}[\text{estimate} = \text{truth}]$) is near zero by construction. When a cheap embedding proxy is available, SUPG (embed) achieves full coverage with width 228—comparable to VAQP-CP—at the same n-call LLM budget, confirming that proxy-guided sampling is competitive when a truly cheap proxy exists. Note that the embed-proxy comparison is filter-only; embedding similarity is not directly applicable to group-by classification or top-k scoring without task-specific fine-tuning.

VAQP-EB achieves perfect coverage (1.000) on filter and group-by but at the cost of much wider intervals (445.5 and 1421.6). This illustrates the CP-vs-EB trade-off: EB uses a looser concentration inequality (empirical Bernstein) that guarantees coverage under weaker assumptions but sacrifices precision. CP with FPC represents the sweet spot for our setting.

SUPG drives filter coverage to 1.0, but at the cost of dramatically wider intervals (430 vs. 193) and—crucially—$N + n$ total LLM calls versus just n for VAQP-CP. On top-k, SUPG *under-covers* at 0.807, suggesting its Hoeffding bounds are not well-suited to rank estimation in this setting.

This clarifies the paper's position: VAQP is not about squeezing margins from a proxy-guided sampler; it is about obtaining a practical, calibrated interval with a simple implementation, minimal LLM cost, and a transparent failure mode.

Table 9: Average RQ2 results across all dataset–model–sample-size configurations. Coverage is measured against the cheap model's full-population output. VAQP-CP achieves the best coverage–width trade-off. VAQP-EB is conservative with perfect coverage but wider intervals.

Operator	Method	Avg. cov	Avg. width	LLM calls
Filter	VAQP-CP	**0.956**	**192.6**	n
Filter	VAQP-EB	1.000	445.5	n
Filter	Naive	0.942	184.9	n
Filter	SUPG (LLM proxy)	1.000	430.4	$N+n$
Filter	SUPG (embed proxy)	1.000	228.0	n
Filter	ABae	0.959	194.7	n
Filter	LOTUS†	0.028	0.0	N
Group-by	VAQP-CP	**0.969**	**632.7**	n
Group-by	VAQP-EB	1.000	1421.6	n
Group-by	Naive	0.928	609.3	n
Group-by	ABae	0.956	635.8	n
Group-by	LOTUS†	0.000	0.0	N
Top-k	VAQP-CP	**0.987**	**121.3**	n
Top-k	Naive	0.950	122.6	n
Top-k	SUPG	0.807	70.9	$N+n$
Top-k	LOTUS†	0.315	0.0	N

†Point estimate, no CI; coverage = $\mathbf{1}[\text{estimate} = \text{truth}]$. Included as cost upper bound.

Figure 5 visualizes the coverage comparison across all three operators. VAQP-CP (blue) consistently exceeds the 0.95 target across all operators, while Naive (red) falls short on group-by. Methods that achieve perfect coverage (VAQP-EB, SUPG) do so at the cost of substantially wider intervals (Table 9). LOTUS, which provides no CI, collapses to near-zero measured coverage.

Group-by analysis. The group-by operator is the most challenging for CI construction because simultaneous coverage over $|C| = 5$ classes requires tighter per-class bounds. Despite this, VAQP-CP achieves 0.969 average joint coverage—well above the 0.95 target—while Naive falls to 0.928. The Bonferroni correction is doing its job: it sacrifices 3.8% in width to gain 4.1 pp in coverage. ABae performs comparably to VAQP-CP (0.956), suggesting that the bootstrap can also handle the simultaneous coverage requirement when n is large enough.

Top-k analysis. On top-k, VAQP-CP achieves 0.987 coverage with width 121.3, while Naive achieves 0.950 with width 122.6. Notably, VAQP-CP is both better-calibrated *and* slightly narrower than Naive on this operator. SUPG under-covers at 0.807—its Hoeffding bounds are designed for binary recall guarantees and do not transfer well to rank estimation. LOTUS achieves only 0.315 coverage because its point estimate (with no CI) rarely equals the exact population rank count.

6.3 RQ3: Does the audit prevent silent failure?

The synthetic stress test (Table 10) is the clearest argument for the audit layer. As flip probability increases, p_L decreases monotonically from 0.928 to 0.372, and SAFE rate drops from 1.0 to 0.0. The interval remains perfectly calibrated with respect to the corrupted model's own full-data semantics: coverage versus noisy full execution stays at 1.0 across the entire sweep. The key systems point is that Naive and ABae provide no such visibility—they keep returning seemingly legitimate intervals while ground-truth coverage drops to 0.0.

Table 10: Stress test under synthetic label corruption. The audit lower bound p_L decreases monotonically with flip probability; SAFE rate tracks it. Naive and ABae silently collapse. Coverage versus the corrupted model's own full-execution semantics (noisy cov) stays 1.0 for VAQP throughout.

Flip prob	p_L	SAFE rate	Cov vs GT: VAQP	Cov vs GT: Naive	Noisy cov (VAQP)
0.00	0.928	1.000	1.000	0.960	1.000
0.05	0.853	0.673	1.000	0.920	1.000
0.10	0.788	0.267	1.000	0.783	1.000
0.15	0.729	0.040	0.993	0.553	1.000
0.20	0.677	0.010	0.947	0.347	1.000
0.30	0.556	0.000	0.660	0.050	1.000
0.50	0.372	0.000	0.033	0.000	1.000

Figure 6 visualizes this trend. The VAQP noisy-coverage line stays flat at 1.0, confirming correct calibration against the model's own semantics. Meanwhile, both VAQP and Naive ground-truth coverage degrade, but only VAQP's audit (p_L curve) provides an early warning signal.

Real-model case study. Table 11 shows three regimes. *FEVER*: low agreement (0.741–0.810), audit flags ≥ 99% as UNSAFE, yet GT coverage stays ≥ 0.895—the wide CI absorbs the discrepancy. *Amazon*: moderate agreement (0.928–0.932), SAFE rate 0.220–0.465; GT

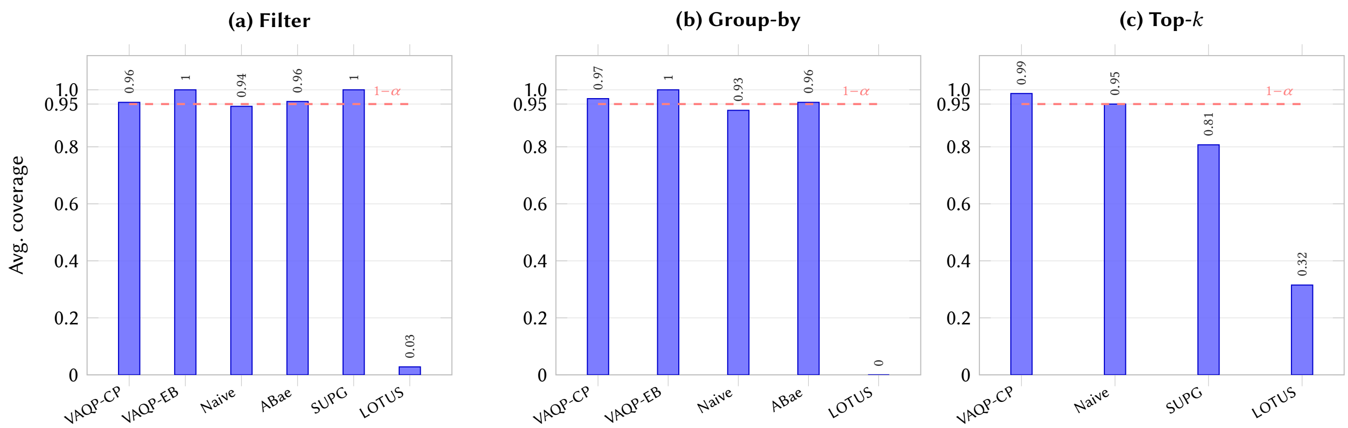

Figure 5: RQ2 coverage comparison across three operators (against cheap-model full execution). Each bar shows average empirical coverage over all dataset–model–sample-size configurations. The dashed red line marks the $1-\alpha = 0.95$ target. VAQP-CP exceeds the target on all operators; Naive under-covers on group-by (0.928); SUPG (LLM proxy) under-covers on top-k (0.807); LOTUS collapses without a CI. Embed-proxy SUPG results are in Table 9.

Table 11: Real-model degradation (detailed). Each row is one (dataset, cheap model, sample size) configuration with `gpt-4o-mini` as oracle. Agreement is measured between the cheap model and oracle on the full population. Coverage is against human-labeled ground truth (200 trials).

Dataset	Cheap model	n	Agreement	SAFE rate	VAQP cov (GT)	Naive cov (GT)
FEVER	deepseek	100	0.810	0.010	0.965	0.950
FEVER	deepseek	200	0.810	0.005	0.960	0.960
FEVER	qwen	100	0.741	0.000	0.955	0.930
FEVER	qwen	200	0.741	0.000	0.895	0.895
Amazon	deepseek	100	0.928	0.220	0.945	0.940
Amazon	deepseek	200	0.928	0.400	0.930	0.925
Amazon	qwen	100	0.932	0.235	0.935	0.915
Amazon	qwen	200	0.932	0.465	0.850	0.840
Yelp	deepseek	100	0.896	0.135	0.935	0.930
Yelp	deepseek	200	0.896	0.115	0.955	0.945
Yelp	qwen	100	0.944	0.350	0.725	0.725
Yelp	qwen	200	0.944	0.610	0.455	0.415

coverage tracks SAFE rate. *Yelp* (the failure case): agreement 0.944, SAFE rate 0.610, yet GT coverage drops to 0.455—both models share a systematic bias that the audit cannot detect. This confirms the guarantee boundary: the audit measures cheap–oracle consistency, not truth. We recommend complementing the audit with periodic human evaluation, especially on new domains.

6.4 RQ4: Pareto trade-offs and scalability

Figure 7 plots coverage vs. LLM calls for 48 filter configurations. VAQP-CP achieves 0.962 coverage at 175 calls—a 2.0 pp improvement over Naive at identical cost; SUPG and Full-LLM reach 1.0 at 6.7× higher cost.

Scalability. VAQP-CP maintains coverage 0.925–0.980 as N grows to 5,000; relative width shrinks from 0.165 to 0.051.

Stratified vs. uniform sampling. Table 13 compares length-stratified sampling (§3.6) against uniform on filter (3 datasets, 2 models, 200 trials). Neyman allocation reduces CI width by 3.2%–5.1% at no extra LLM cost.

Table 12: Scalability on Amazon filter ($n = 200$, 200 trials). $w/N = O(\sqrt{n}/N)$.

N	Model	Method	Coverage	Rel. width
500	deepseek	VAQP-CP	0.980	0.165
500	deepseek	Naive	0.965	0.158
1 000	deepseek	VAQP-CP	0.950	0.113
1 000	deepseek	Naive	0.960	0.110
2 000	deepseek	VAQP-CP	0.925	0.081
2 000	deepseek	Naive	0.945	0.078
5 000	deepseek	VAQP-CP	0.945	0.051
5 000	deepseek	Naive	0.930	0.050
500	gpt-4o-mini	VAQP-CP	0.960	0.172
1 000	gpt-4o-mini	VAQP-CP	0.945	0.116

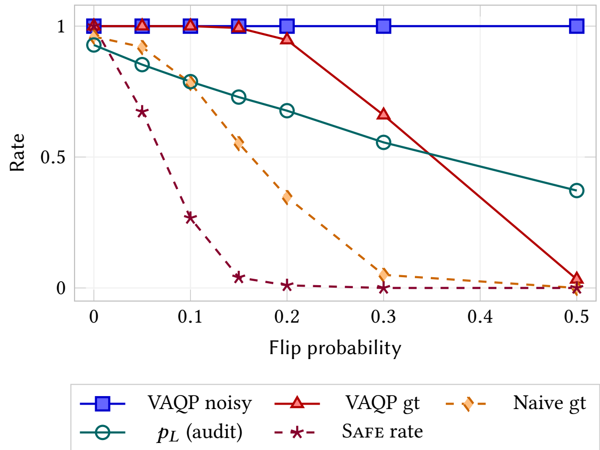

Figure 6: Stress test: as flip probability increases, VAQP's coverage against the noisy model's own semantics remains perfect (1.0, blue), while ground-truth coverage degrades for both VAQP (red) and Naive (orange). The audit lower bound p_L (teal) provides an early warning signal that tracks the degradation.

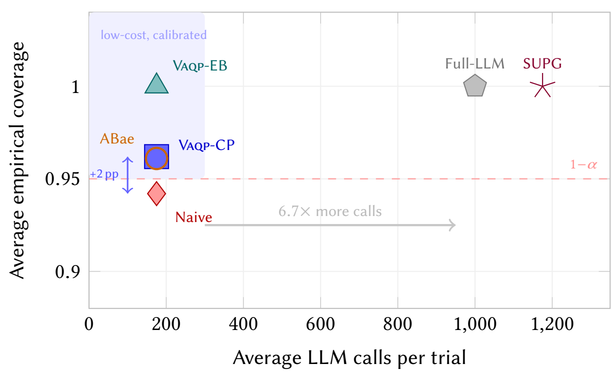

Figure 7: Pareto frontier: coverage vs. LLM calls (48 filter configs). VAQP-CP achieves 0.962 at 175 calls (+2 pp over Naive); SUPG/Full-LLM reach 1.0 at 6–7× cost.

Table 13: Stratified vs. uniform sampling (filter, $N = 1{,}000$, 200 trials). Width reduction is relative to Uniform-CP.

Dataset	n	Mean CI width		Δw
		Uniform	Stratified	
FEVER	100	172.8	164.3	−4.9%
FEVER	200	113.5	109.9	−3.2%
Amazon	100	176.6	167.6	−5.1%
Amazon	200	117.1	112.3	−4.1%
Yelp	100	191.6	183.4	−4.3%
Yelp	200	127.0	122.2	−3.8%

7 RELATED WORK

Classical AQP.. Online aggregation [12], AQUA [1], BlinkDB [2], VerdictDB [19], and AQP++ [21] approximate numeric aggregates via sampling, synopses, or scrambled tables. As analyzed in Table 1, all assume free, deterministic predicates—LLM operators violate both. VAQP fills these gaps with exact non-asymptotic CIs, fresh-sample execution, and a verification layer.

LLM semantic query systems. LOTUS [20] exposes semantic operators with per-item recall targets but no aggregate CIs; Sema [22], Abacus [25], and Cortex AISQL [17] optimize execution without formal statistical bounds. SUPG [16], InQuest [24], NoScope [15], and BlazeIt [14] use cheap proxy scores for efficient querying over ML-labeled data [6, 10]. VAQP extends this line to group-by, top-k, and text-domain LLM operators with an audit layer; unlike SUPG/InQuest, it does not require pre-scoring all N documents. Conformal prediction [4, 29] provides per-instance guarantees but not aggregate-level CIs.

8 DISCUSSION AND LIMITATIONS

Limitations. The audit certifies cheap–oracle consistency; correlation with human truth depends on oracle quality (Yelp: agreement 0.944, GT coverage 0.455); we recommend selecting the oracle from a different model family. The top-k CI is a population-rank interval, not a confidence set over item identities.

Conclusion. Across 180 configurations, VAQP-CP meets the 0.95 target on all operators (min 0.950), improves coverage by 1.4–4.1 pp over Naive with ∼4% wider intervals, reduces LLM calls by 80–96%, and length-stratified sampling further tightens CIs by 3–5% at zero extra cost. Code and response cache will be released under MIT license.

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